## A Task formulation and data

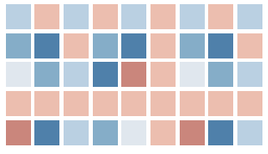
One perturbation target per sample; K562 response vectors provide supervision

**GenePT embedding (3,072-D)**
target-gene representation

**UCE embedding (5,120-D)**
target-gene representation

**Matched targert**
ID / row alignment

**K562 response**
ΔY: 1,950 × 8,563



Observed ΔY

Input: target-gene embeddings
output: predicted transcriptomic response

## B Gated Direct: model computation

Feature-wise fusion of two fixed pretrained gene representations

**GenePT → Linear / LN / GELU**

**UCE → Linear / LN / GELU**

**hG ∈ R^w**

**hU ∈ R^w**

**Concat [hG; hU]**
Linear(2w → w) + Sigmoid

**g ⊙ hG + (1 − g) ⊙ hU**

**Dropout → Linear → GELU → Dropout → Linear**

**ŷ = ΔŶ**

## C Controlled architectural ablations

All four variants use the same direct-response MLP head

| | |
|---|---|
| **Gated fusion** | **g ⊙ hG + (1 − g) ⊙ hU** Learned feature-wise gate |
| **No-gate fusion** | **0.5 × (hG + hU)** Equal-weight average |
| **GenePT-only** | **hG** GenePT projection only |
| **UCE-only** | **hU** UCE projection only |

Direct-response benchmark: no Ridge residual is added to these four model outputs.

## D Leakage-aware evaluation

Hyperparameter selection on K562 only; external evaluation with frozen weights

**K562 targets**
outer training split

**Inner validation**
K562-only tuning

**Selected model**
refit + freeze weights

**K562 held-out**
unseen perturbations

**RPE1 held-out**
matched perturbations

**HepG2 / Jurkat**
seen vs held-out

All evaluation uses K562-trained weights; no external-cell-line refitting.